Solar energy - 47202 & 47214

resource, solar thermal and photovoltaics

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Outline

Solar resource
- The sunlight in space
- The sunlight on Earth's surface

Conversion technologies
- Solar thermal energy

Break

Conversion technologies (cont.)
- Photovoltaics

Feedback

Activities

Learning objectives

Solar energy

By the end of this lecture, you should be able to

- Identify true or false statements related to solar energy resources, solar thermal energy efficiency, and solar PV cells;
- Calculate technical parameters related to solar energy resources, solar thermal energy, and photovoltaics;
- Calculate cross-disciplinary parameters related to thermodynamics and the capacity factor of solar energy.

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The sun

from fusion reactions to radiation

Fussion reactions heat particles up to high temperatures, which dissipate energy as thermal radiation ($\dot{Q}_{rad}^{out}$)

$$\dot{Q}_{rad}^{out} = \epsilon \sigma A T^4 \qquad (1)$$

- ϵ: emissivity [-]
- $\sigma \approx 5.67 \cdot 10^{-8}$ $[W/m^2K^4]$
- A: Area of emitting body $[m^2]$
- T: Absolute temperature [K]

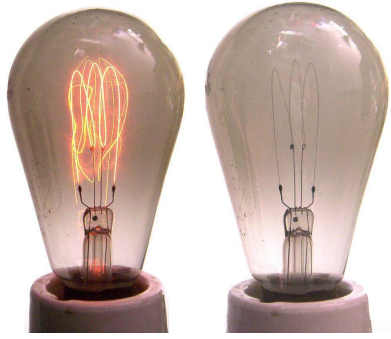


Figure: Incandescent light bulb [1]

Irradiance (I)

and the solar constant

The average annual solar irradiance at the top of earth (solar constant) is 1,361 $\mathrm{W/m^2}$

Definition

$$I = \frac{\dot{Q}^{in}_{rad}}{A} \qquad (2)$$

- $\dot{Q}^{in}_{rad}$: incoming thermal radiation [W]
- A: incoming area $[\mathrm{m^2}]$

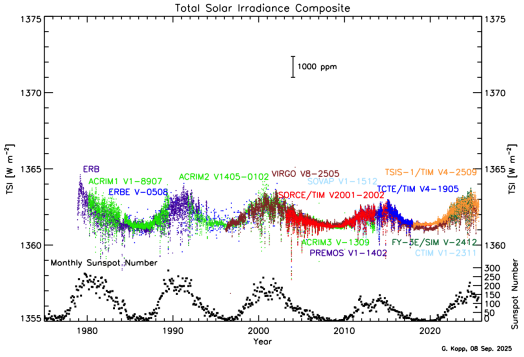


Figure: Compilation of measured total solar irradiance

Activity

Total solar energy resource

Problem

If the solar constant is 1,361 $\mathrm{W/m^2}$ and the Earth's diameter is 12,756 km, how much thermal radiation does the Earth receive from the Sun?

a. *175 TW*
b. *700 TW*
c. *175000 TW*
d. *700000 TW*
e. *Above 700000 TW*

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The earth's radiative energy balance

the impact of the atmosphere on sunlight

Three things can happen when light reaches a body:

1. Absorb ($\dot{E}_{abs}$)
2. Reflect ($\dot{E}_{ref}$)
3. Transmit ($\dot{E}_{trans}$)

$$\dot{Q}^{in}_{rad} = \dot{E}_{abs} + \dot{E}_{ref} + \dot{E}_{trans} \quad (3)$$

The atmosphere is essential to maintain Earth's temperature

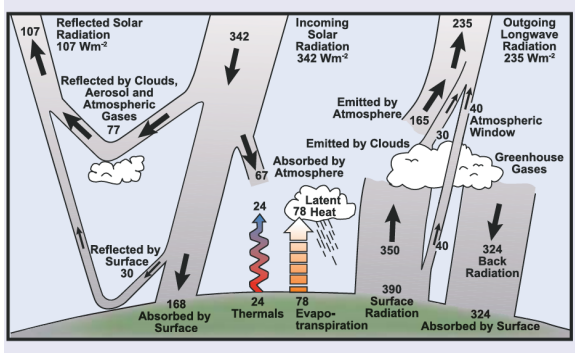


Figure: Earth's radiative balance (24h average) [2]

Irradiance classifications

Type, orientation, and position

Definition

- *Direct* irradiance (I_{direct}) is the sunlight derived directly from the sun.
- *Diffuse* irradiance ($I_{diffuse}$) is derived from the scattering of sunlight.
- *Global* irradiance (I_{global}) is the sum of direct and diffuse sunlight.

$$I_{global} = I_{direct} + I_{diffuse} \quad (4)$$

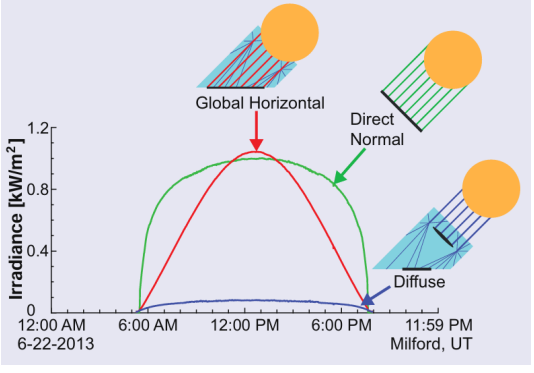


Figure: Irradiance Profiles at the Earth's Surface on a Cloudless Day [2]

Irradiance classifications

Type, orientation, and position

Definition

- Normal (I^{normal}): surface is perpendicular to sunlight
- Horizontal ($I^{horizontal}$): surface is parallel to the ground

For direct irradiance:

$$I_{direct}^{horizontal} = I_{direct}^{normal} cos(z) \qquad (4)$$

In which z is the zenith angle.

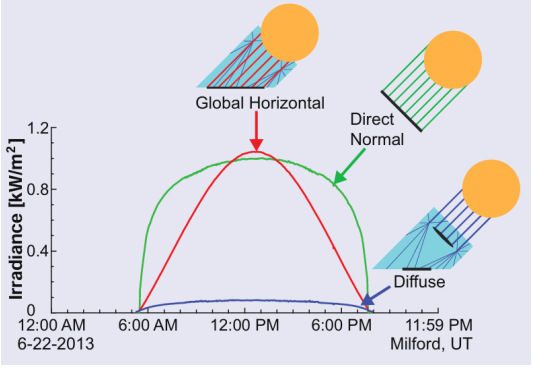


Figure: Irradiance Profiles at the Earth's Surface on a Cloudless Day [2]

Solar irradiation ($\bar{I}$)

irradiance over a specific period of time

Definition

$$\bar{I} \approx \sum_i I_i \Delta t_i \qquad (5)$$

- I_i: irradiance over period i $[\mathrm{kW/m^2}]$
- Δt_i: time period i $[h]$

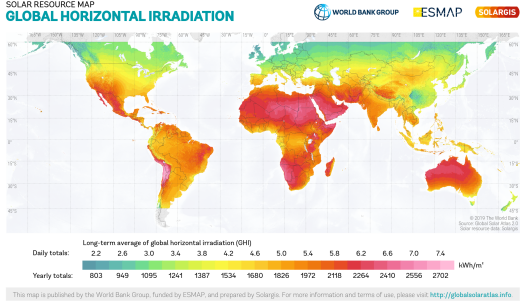


Figure: Average global horizontal irradiation [3]

Activity

Global solar atlas

Problem

Calculate the ratio between the global horizontal irradiation (GHI) of your hometown and Copenhagen using data from the Global Solar Atlas.

$$\frac{GHI_{hometown}}{GHI_{CPH}} = ?$$

We will calculate the class average through Vevox

Intermittence

predictable and unpredictable factors

Solar irradiance have predictable changes during the day and year because of the Earth's tilted orbit and rotation.

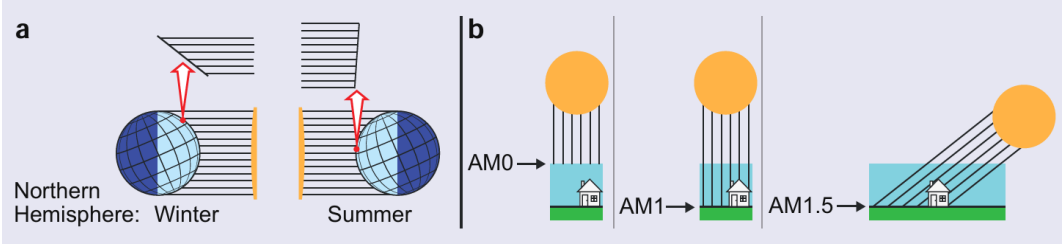


Figure: Incident Solar Radiation, Effect of Seasonal Variation (a), and Effect of Atmospheric Attenuation (b) [2]

Intermittence

predictable and unpredictable factors

Solar irradiance have (mostly) unpredictable changes during the day due to the weather conditions (i.e., clouds, shadows)



Figure: Clouds block direct sunlight [5]

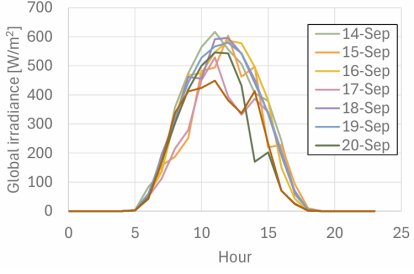


Figure: Daily irradiance over a week in DK

The sunlight on Earth's surface

Geographical distribution

evenly distributed, but with demand mismatches

Highest supply: Africa and Oceania

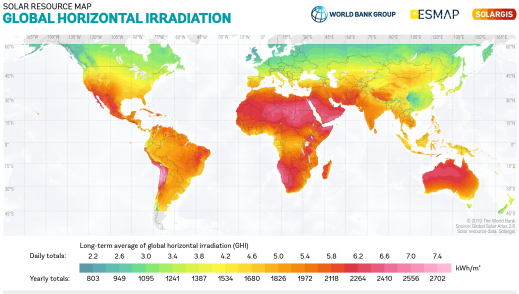


Figure: Average solar irradiance [3]

Highest demand: US, EU, and Asia

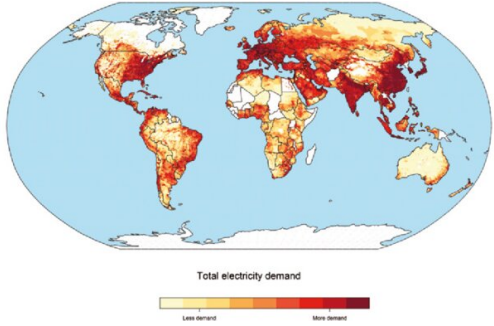


Figure: Electricity demand by population [4]

Activity

Geographical impacts on solar energy plant designs

Problem

Discuss in pairs: One satellite image is from Denmark and another is from Chile. Which is which and why?





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Converting sunlight energy

Most energy sources are derived from the sun

In this lecture, we will focus only on solar thermal energy and photovoltaics

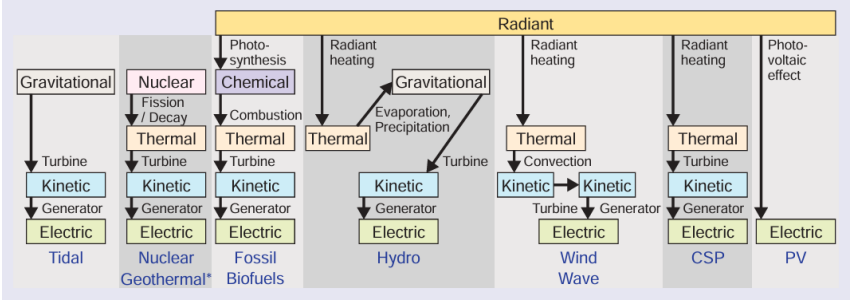


Figure: Forms of Energy and Energy Conversion Processes [2]

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Activities

Solar collectors

using sunlight to heat a fluid

Instead of using fuels or electricity, we can use sunlight for heating and drying.



Figure: A solar collector heating water [6]

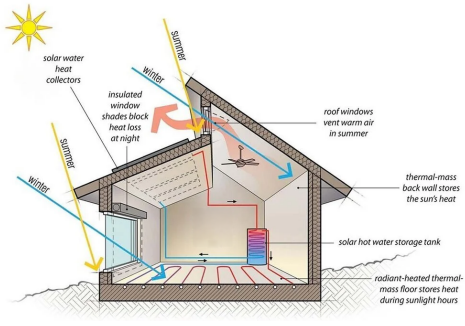


Figure: Solar passive heating [7]

Solar power plant

from heat to electricity

A solar power plant consists of:

1. Solar collector: Linear, disk-engine, or tower
2. Collector fluid loop: water, oil, or molten salt
3. Steam power cycle (Rankine)

Highlight

Instead of burning a fuel, a solar power plant collects energy from sunlight to generate electricity

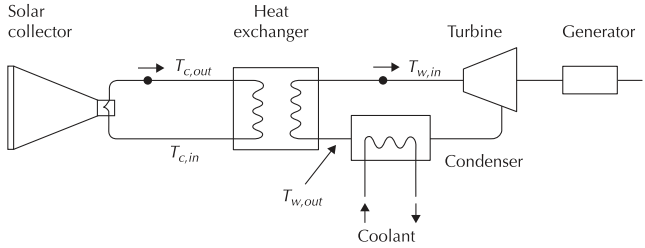


Figure: A generic solar power plant [10]

Solar power plant

from heat to electricity

η	CF	CAPEX	LCOE
11-22%	22%	2418 $/kW	115$/MWh



Figure: Solar tower collector [9]

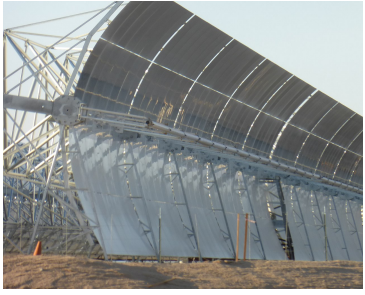


Figure: Linear solar collector [9]



Figure: Disk/engine collector [9]

Solar collector engineering

trade-offs between heat collection and conversion efficiencies

Radiation losses: $\uparrow T \implies \uparrow Q_{losses}$

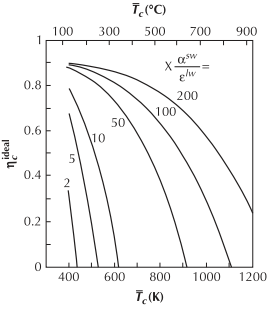


Figure: Ideal collector efficiency [10]

($\downarrow T$ and $\downarrow \eta_{cycle}$) vs. ($\uparrow T$ and $\uparrow Q_{losses}$)

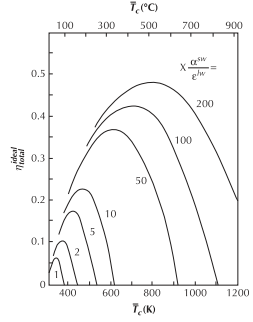


Figure: Ideal CSP efficiency vs. T_c [10]

Solar collector engineering

trade-offs between heat collection and conversion efficiencies

($\downarrow T$ and $\downarrow \eta_{cycle}$) vs. ($\uparrow T$ and $\uparrow Q_{losses}$)

Engineering solutions

- Enhance the collector material absorptance (α) vs. emissivity (ϵ) properties

$$\uparrow \frac{\alpha}{\epsilon}$$

- Increase the ratio of solar capture area ($A_{capture}$) vs. collector area ($A_{collector}$) (i.e., concentration factor, X)

$$\uparrow \frac{A_{capture.}}{A_{collector}} = \uparrow X$$

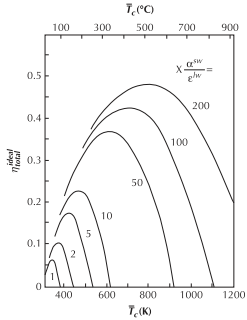


Figure: Ideal CSP efficiency vs. T_c [10]

Advantages

heat storage and simple technology

The main advantage of CSP (compared to PV) is the integration with heat storage to deliver power during the night.

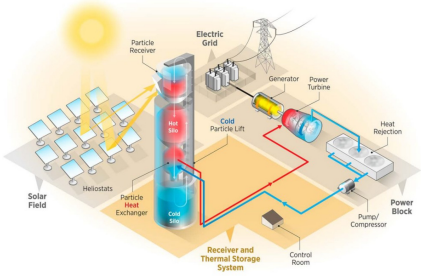


Figure: CSP with thermal storage [8]

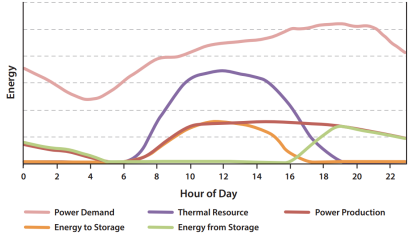


Figure: Use of Thermal Energy Storage in a CSP Plant [2]

Activity

CSP efficiencies

Problem

Discuss in pairs: Which CSP technology should be more efficient and why?



Figure: Solar tower collector [9]

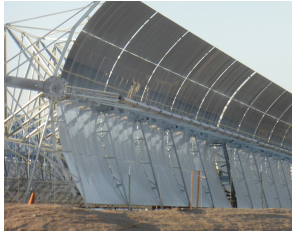


Figure: Linear solar collector [9]



Figure: Disk/engine collector [9]

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Short break?

Should we have a short break and return in 10 minutes?

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Photons

and the photoelectric effect

Light travels at constant speed (c):

$$c = f\lambda \qquad (6)$$

Light transfers energy (E) in discrete packets called photons.

$$E = hf = \frac{hc}{\lambda} \qquad (7)$$

c	299,792,458 [m/s]
f	Frequency [Hz]
λ	Wavelength [m]
E	Energy [J]
h	$6.626 \cdot 10^{-34}$ [J/Hz]

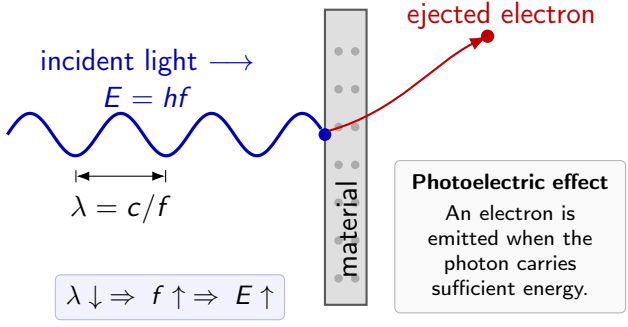


Figure: Photoelectric effect and wave equations

Semiconductors

Conduction, valence, and band gap

Semiconductors are intermediary materials between metals and insulators with a band gap small enough for electrons to cross by absorbing energy (as heat or photons).

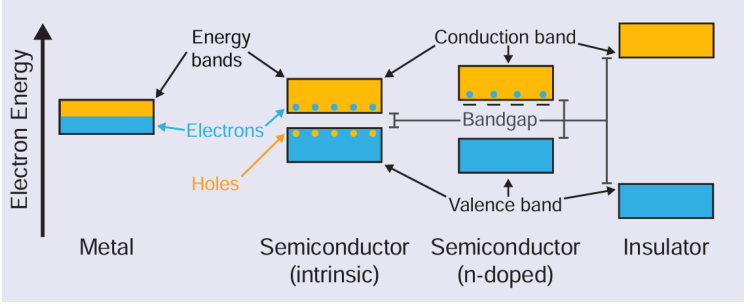


Figure: Energy band structure of different materials [2]

Semiconductors

Conduction, valence, and band gap

Adding impurities can create extra free charges (i.e., electrons or "holes") which improve the conductivity of semiconductors.

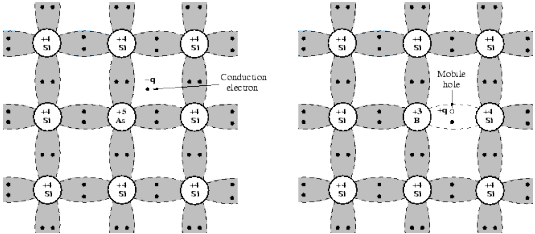


Figure: P-type and n-type doping [11]

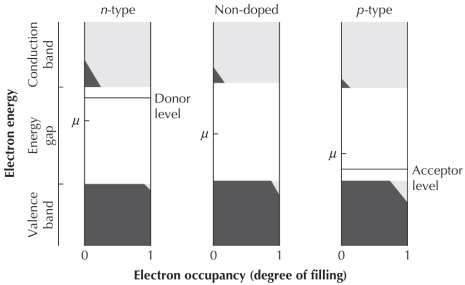


Figure: Doping effect in energy bands [10]

P-n junction

An one-way lane for electrons

By joining p-type and an n-type semiconductors, we can create an internal electrical field in the material that separates charges in a particular direction.

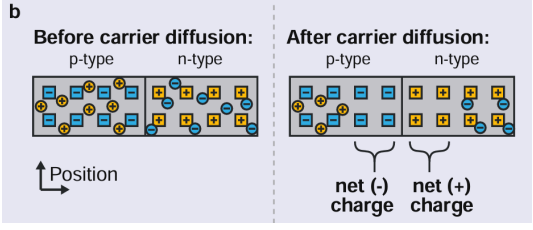


Figure: Internal electric field in pn-junction [2]

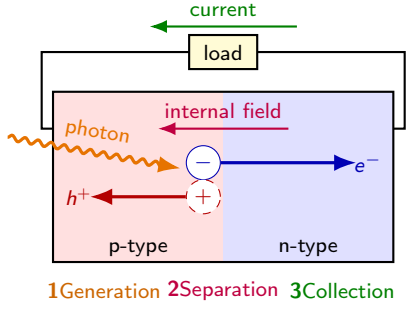


Figure: PV diagram and physical steps

P-n junction

An one-way lane for electrons

From light to electricity

1. Charge generation
2. Charge separation
3. Charge collection

$$E_\gamma \geq E_g$$

Photons below the band gap E_g are not absorbed. Energy above E_g is mainly lost as heat.

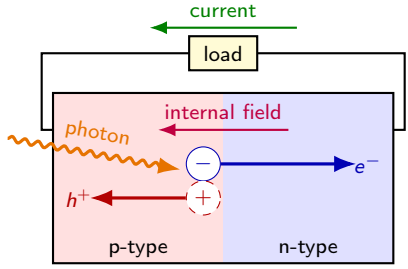


Figure: PV diagram and physical steps

Activity

Band gap limitation

Problem

If the band gap energy for silicon is 1.12 eV, what is the maximum wavelength in μm a photon can have to be able to create an electron-hole pair? Remember that:

$$\lambda\,[m] = \frac{c\,[m/s]}{f\,[1/s]}$$

$$E\,[J] = h\,[J/Hz] \cdot f\,[Hz]$$

In which, $\mathrm{c} = 3 \cdot 10^{8}$ *m/s,* $\mathrm{h} = 6.626 \cdot 10^{-34}$ *J/Hz, and* $1\ \mathrm{eV} = 1.602176634 \cdot 10^{-19}$ *J*

Energy efficiency

How high can we go?

Shockley-Queisser limit[a]: The efficiency limit of a single-junction cell is 33%, but multi-junction cells can reach up to 68%.

[a] As it assumes the solar spectrum at the Earth surface, the limit will be different for extraterrestrial applications

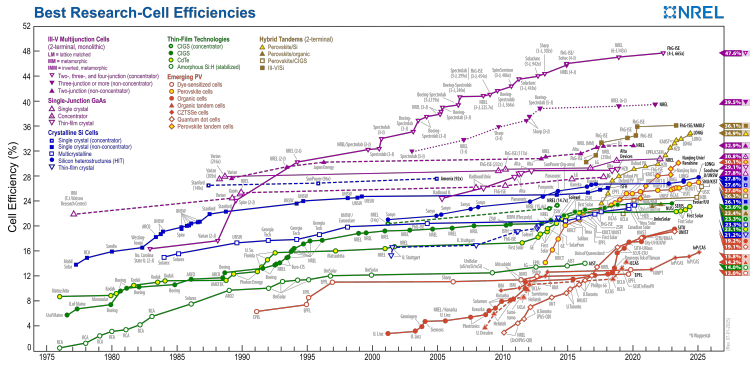


Figure: Best research cell efficiencies [14]

Crystalline silicon cells

the conventional PV cell (90% in 2014)

A highly efficient c-Si cell (PERL) has:

- A very thin and fine metal contact (Ag).
- An inverse pyramid surface with anti-reflection coating.
- Passivation oxide layers to suppress surface recombination.
- A phosphorus-doped n-layer and a boron-doped p-layer.
- Altered degree of doping (p+, n+) [a].

[a] avoided in industry standard cells (PERC - 20% eff.) to save costs.

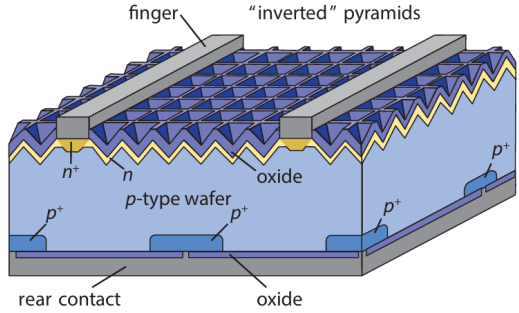


Figure: A mono-crystalline silicon cell with a passivated emitter and a locally diffused rear structure (PERL) - 25% efficiency

Solar photovoltaic plant

from photons to electricity

η	CF	CAPEX	LCOE
20-25%	16%	667 $/kW	44$/MWh



Figure: A conventional PV plant [12]



Figure: A 2-axis following PV [12]



Figure: A vertical PV with plantations [13]

PV characteristic curve

Current vs. voltage relationship

Definition

- Shunt specific current (j_{sc}) [W/m^2]: The current with no voltage load.
- Open-circuit voltage (V_{oc}) [V]: The voltage with no current.

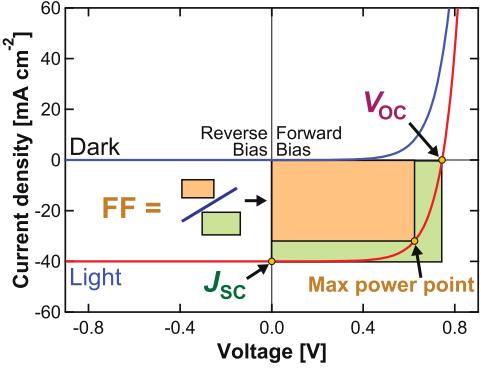


Figure: Characteristic curves with and without sunlight [2]

PV characteristic curve

Current vs. voltage relationship

Definition

Electric power (P):

$$P = V \cdot j \cdot A$$

- P: Electric power [W]
- V: PV cell voltage [V]
- j: PV cell specific current [A/m^2]
- A: PV active area [m^2]

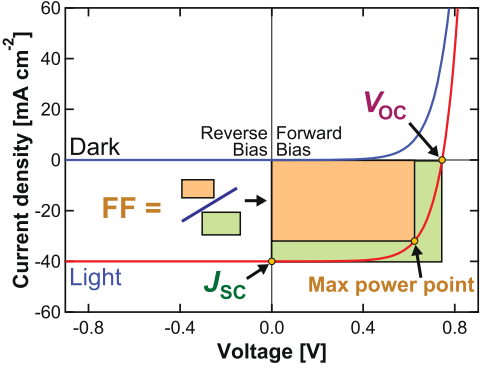


Figure: Characteristic curves with and without sunlight [2]

PV characteristic curve

Current vs. voltage relationship

Definition

Electric efficiency (η):

$$\eta = \frac{P}{I \cdot A} = \frac{V \cdot j}{I}$$

- η: efficiency [-]
- P: Power [W]
- I: Irradiance [W/m^2]
- A: PV active area [m^2]

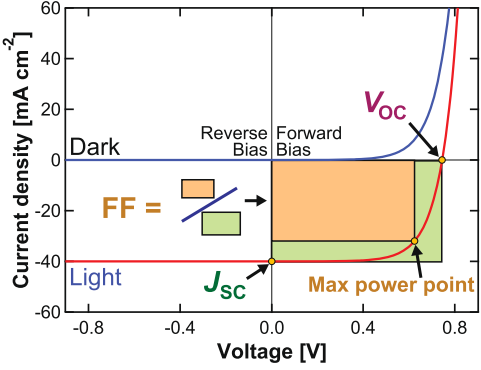


Figure: Characteristic curves with and without sunlight [2]

PV characteristic curve

Current vs. voltage relationship

Definition

Fill factor (FF):

$$FF = \frac{P}{V_{oc} \cdot j_{sc} \cdot A}$$

- P: Power [W]
- V_{oc}: Open current voltage [V]
- j_{sc}: Shunt specific current [A/m^2]
- A: PV active area [m^2]

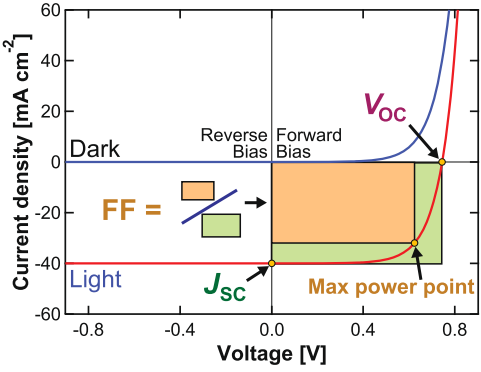


Figure: Characteristic curves with and without sunlight [2]

To keep in mind

Factors influencing PV design and profitability

Degradation

PVs degrade over time (e.g., 0.5%/year)

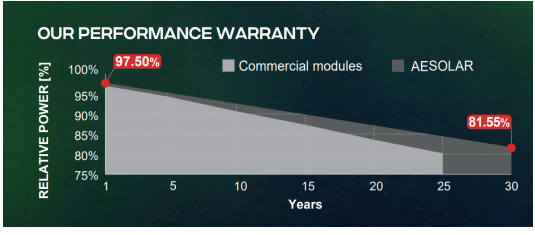


Figure: OEM efficiency warranty [15]

Temperature influence

High temperatures reduce the efficiency (e.g., 0.4-0.5%/°C for Si).

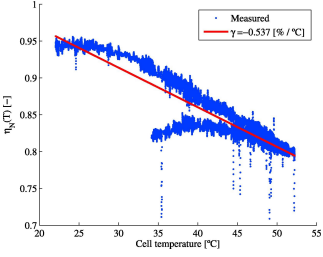


Figure: Normalized η vs. T [16]

Activity

Let's vote!

Problem

Should agricultural land be converted into utility-scale solar parks to accelerate the energy transition?

a *Yes, I'm in favor of it*

b *No, I'm against it*

Note

You will have to defend your position afterwards.

Activity

Let's vote!

Problem

Now, discuss with at least one colleague on: (i) what your position is and (ii) why you choose it ?

Outline

Solar resource
- The sunlight in space
- The sunlight on Earth's surface

Conversion technologies
- Solar thermal energy

Break

Conversion technologies (cont.)
- Photovoltaics

Feedback

Activities

Muddy points and key insights

Feedback for your learning

Here is the chance of you get feedback on your learning.

The quiz will be open only TODAY.

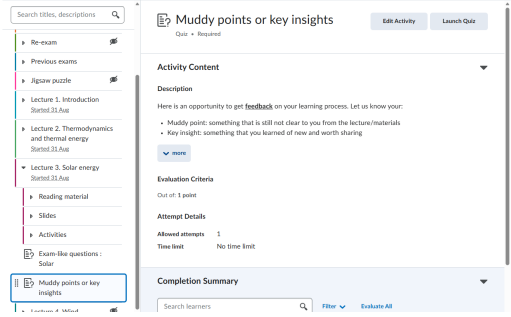


Figure: A screenshot capture of your DTU Learn

Outline

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Feedback

Activities

References I

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